

Glass - 0.98



Smart RVM: Transforming Waste into Wealth

An AI-based object detection framework for label-less, scalable material recycling.



Institution: JAIN (Deemed-to-be University)

Department: Computer Science and Engineering

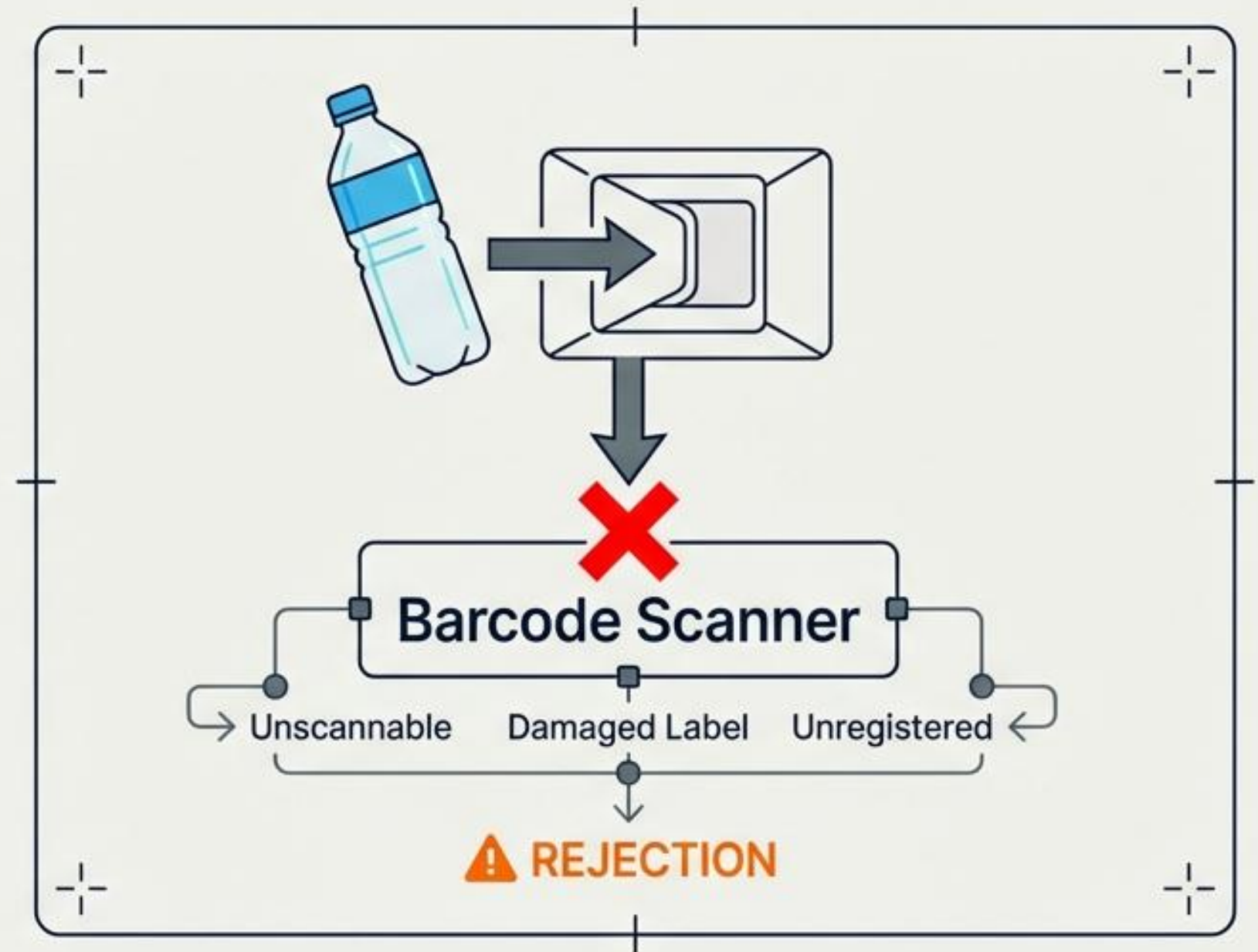
Phase: 6th Semester B.Tech Project Centric Learning

Team: Brayon Tomy , Rohit Shreekumar, S Jayavardhan, Aryan Biswa
Harsh Vardhan Jaiswal

Germany's Pfand scheme proves high-yield recycling works, but its barcode reliance creates a rigid bottleneck.

98%

Traditional Reverse Vending Machines (RVMs) achieve massive return rates for registered, perfectly labelled beverage containers.



Any container without a valid, scannable barcode is automatically rejected.

Millions of valid recyclables are rejected daily because they lack a registered label.

The Excluded Majority



Damaged containers, imported goods, and small-manufacturer items are locked out of automated collection, destroying potential economic and environmental value.

Consumer Friction



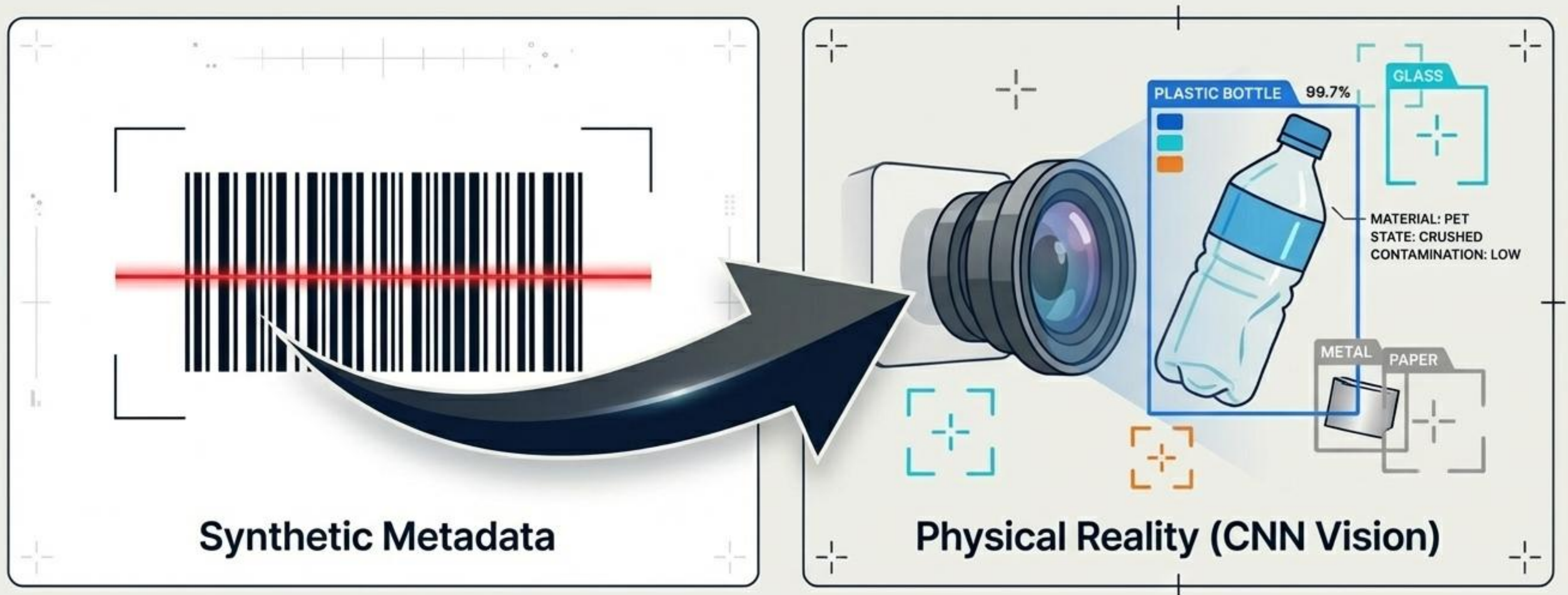
Frequent rejection of perfectly valid items breaks consumer trust, causing frustration and directly reducing long-term recycling participation.

The Hazardous Blindspot



Traditional barcode systems have zero mechanism to flag or separate hazardous waste like batteries or chemicals, leading to dangerous downstream mixing.

The Smart RVM shifts recycling from reading labels to seeing materials.



Plastic

Glass

Paper

Metal

Hazardous

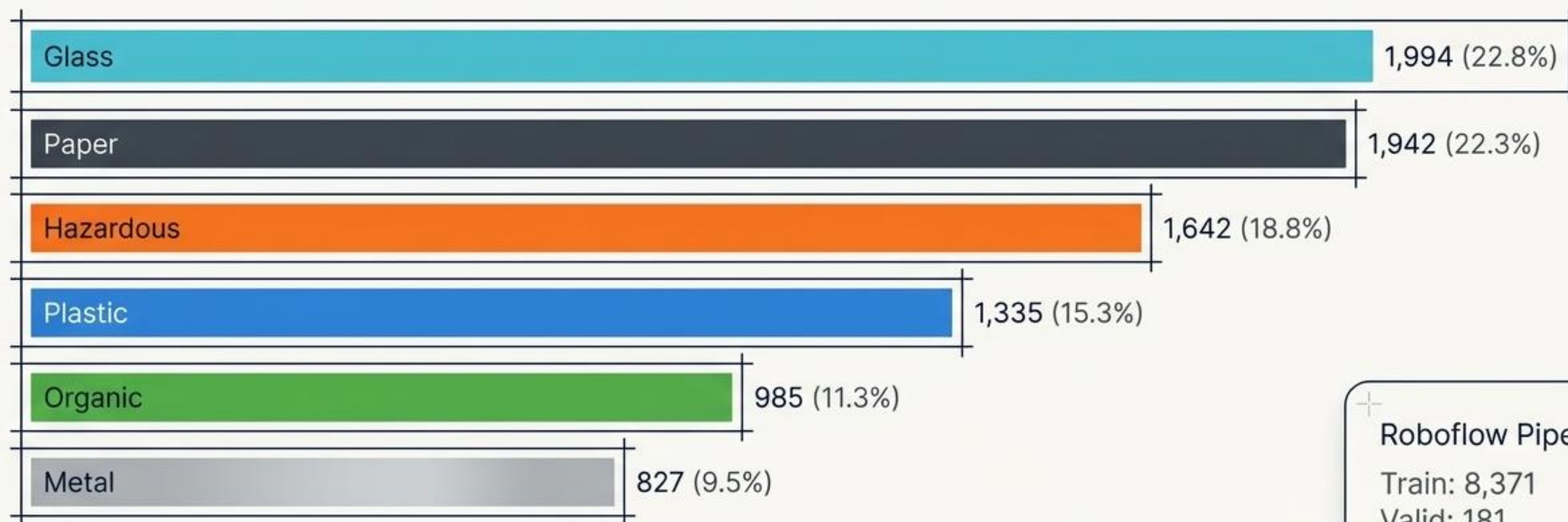
Organic

Diagnostic Comparison: Traditional RVMs vs. AI-Powered Smart RVMs

Feature Dimension	Traditional RVM	Smart RVM
Identification Method	Barcode/RFID Scanning	AI Computer Vision (ResNet34)
Label Condition	Must be pristine	Irrelevant (Label-less detection)
Unlabelled Item Handling	Automatically Rejected	Accepted & Classified
Hazardous Detection	None	Dedicated Classification Stream
Incentive Model	Fixed deposit return	Dynamic Reward Points System

A robust dataset of 8,725 images establishes the baseline for material recognition.

Data Health



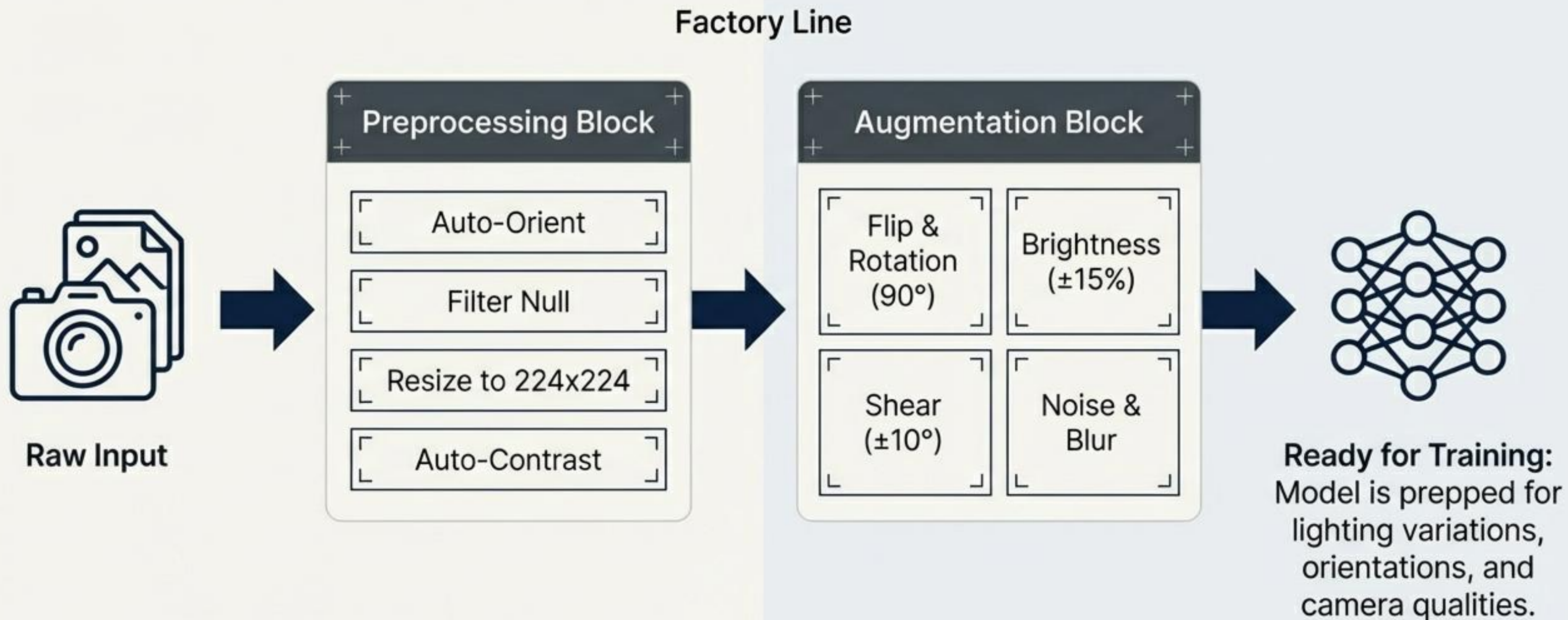
Roboflow Pipeline

Train: 8,371

Valid: 181

Test: 173

Standardized preprocessing and augmentation simulate real-world environmental noise.



Fine-tuning ResNet34 creates a highly accurate, single-class image classifier

Tech Stack Details

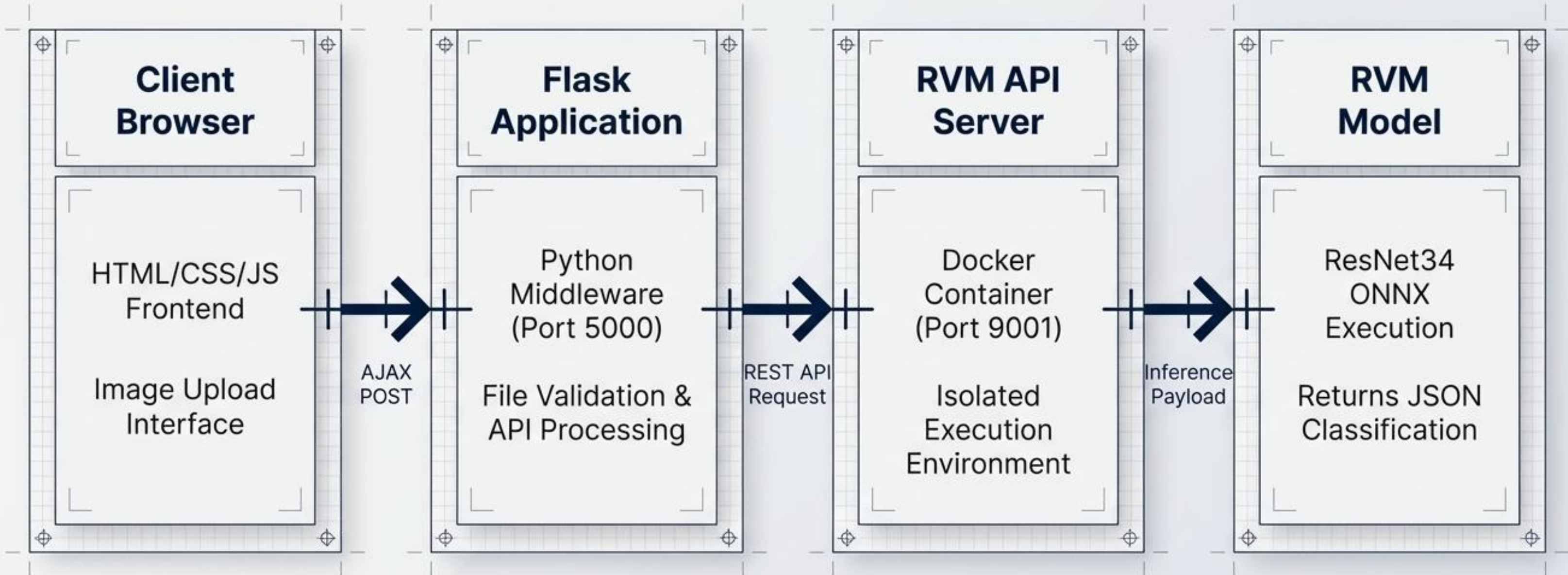
Architecture:	ResNet34 (34-layer residual network)
Starting Checkpoint:	Pre-trained ImageNet weights (Transfer Learning)
Reasoning:	Residual (skip) connections allow gradients to flow effectively, drastically reducing the vanishing gradient problem in deep networks.



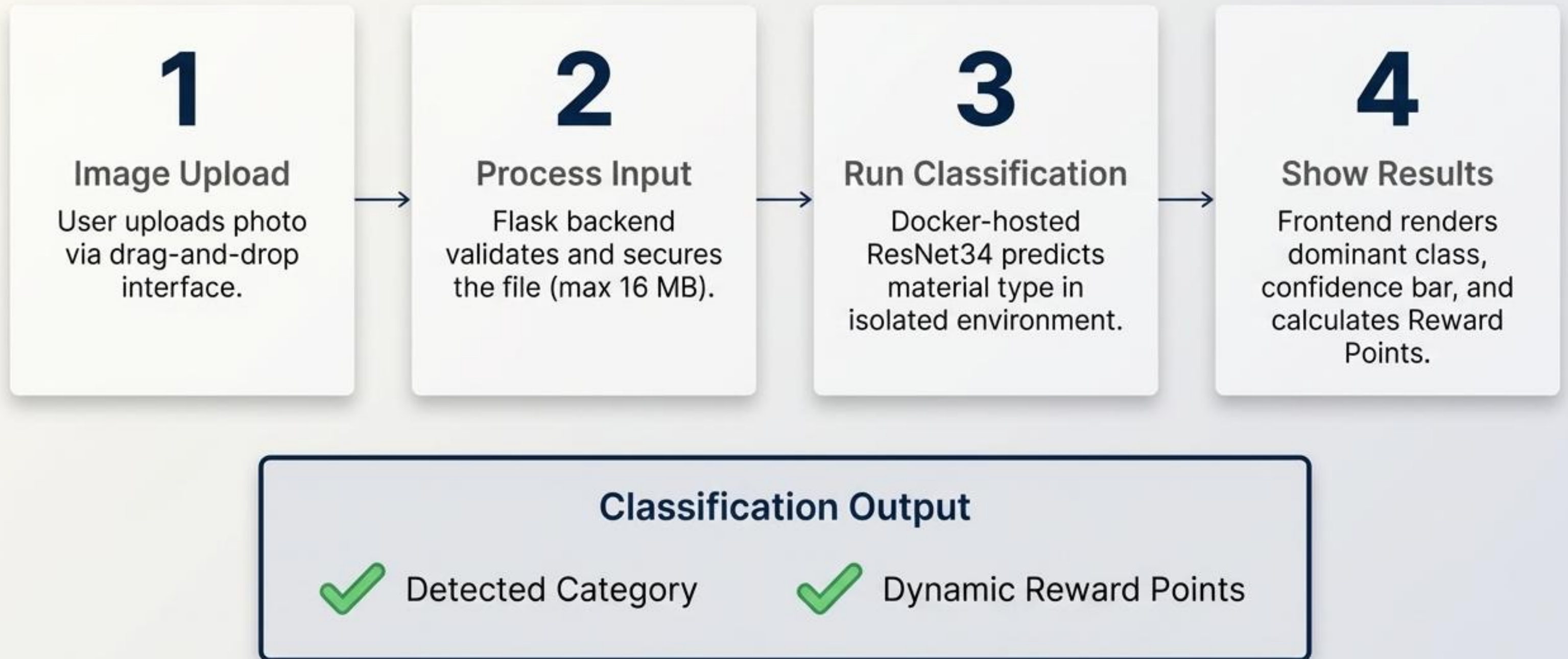
Exported to ONNX runtime format for **optimized** cross-platform execution.

Prediction Inference Time: **0.3 - 0.5s**

A modular, containerized architecture separates the web interface from the ML inference engine



The classification workflow delivers real-time feedback and user rewards in under six seconds.



The RVM V1 model achieves 96.7% validation accuracy with minimal inference latency.

Top-1 Validation Accuracy



96.7%

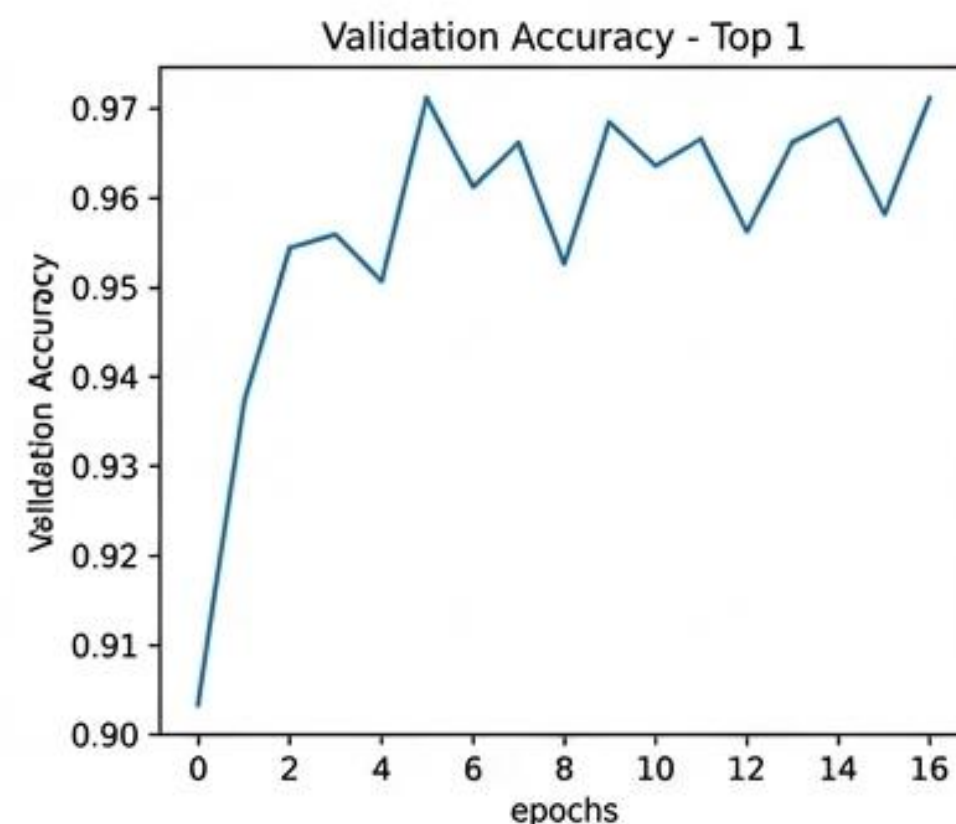
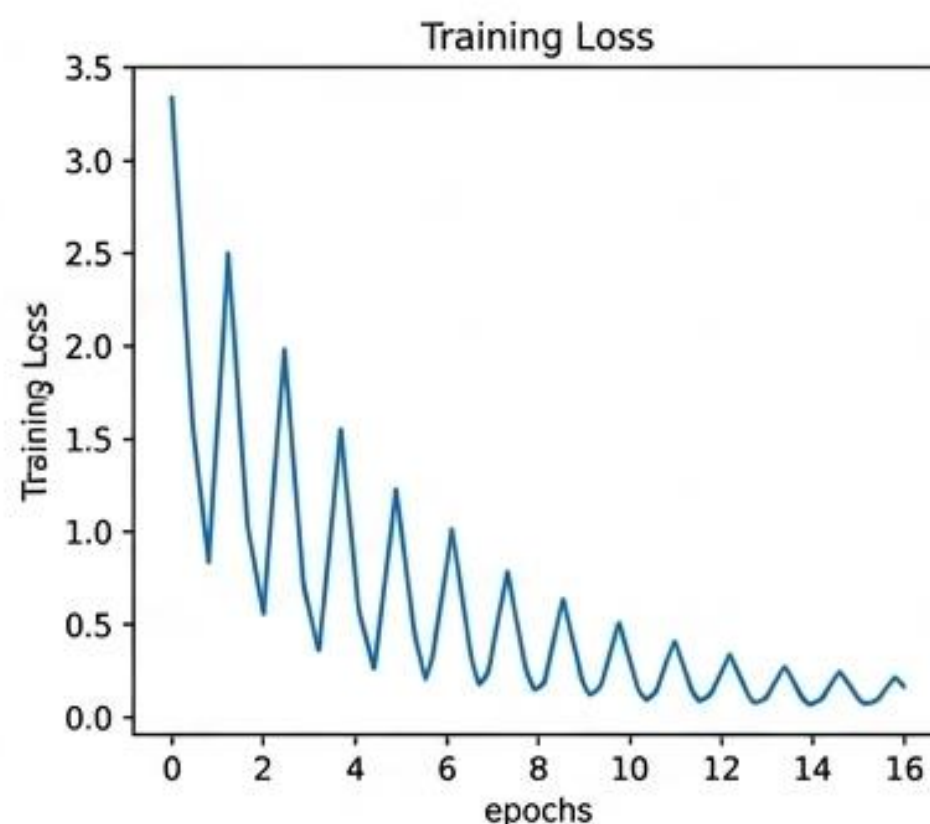
Final Training Loss

0.00002

ML Inference Time

~0.3 - 0.5s

System Health Dashboard



Graph proves model generalized well to unseen data without significant overfitting.

Qualitative testing proves color-invariant classification across diverse, cluttered environments.



Color Invariance (Glass)



High Confidence (Metal)



Real-world Clutter (Plastic)

The user-facing web app prioritizes a frictionless, data-driven experience.

[PREVIEW_MODE]

Real-time Preview

Immediate image validation and visual preview rendered locally before server submission.

[RESPONSIVE_A11Y]

Accessibility

Fully responsive layout supporting desktop and mobile, with an integrated native dark mode toggle.

[STORAGE_API]

Session Persistence

Browser LocalStorage API tracks user scan history and timestamps without requiring login.

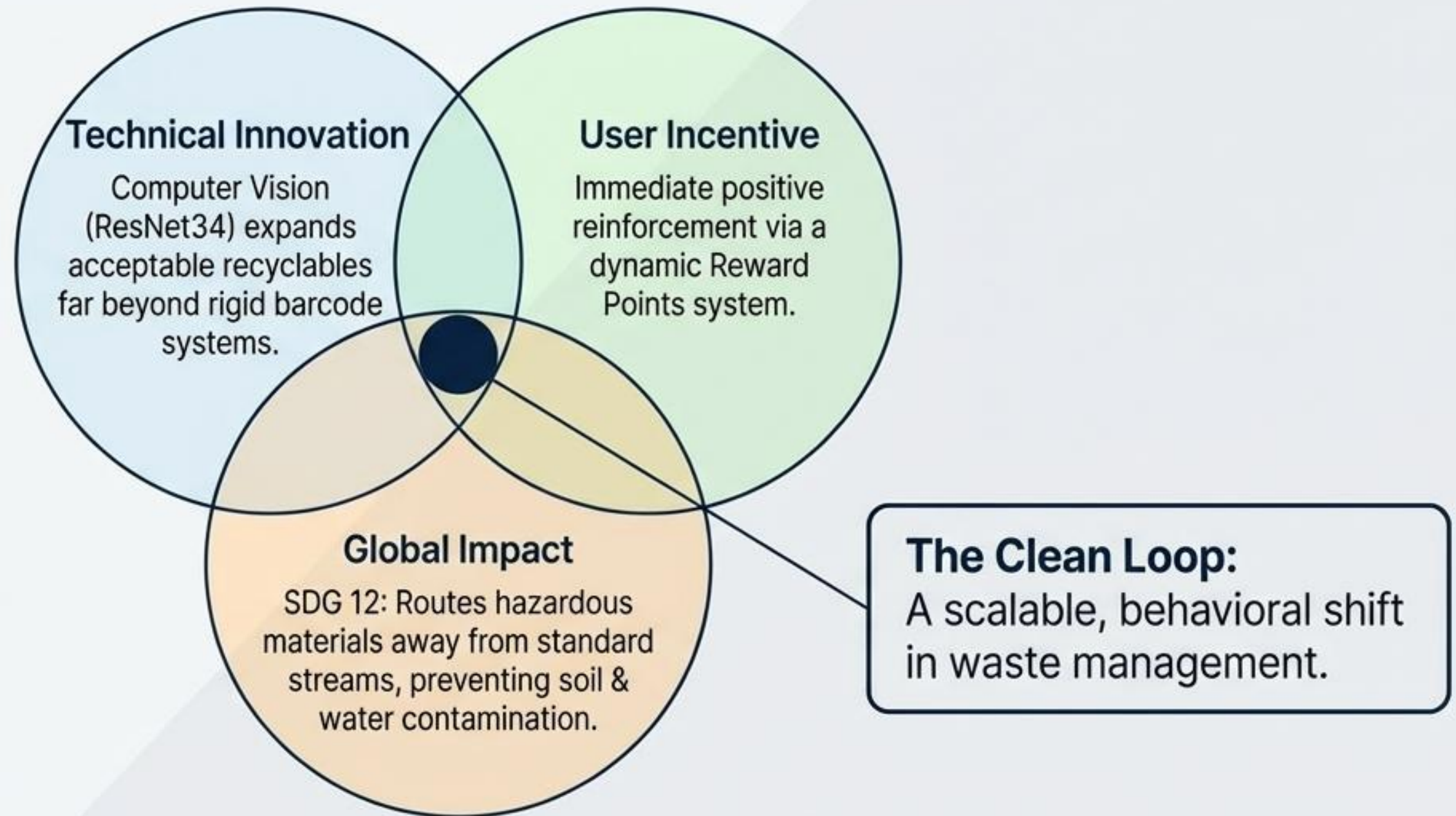
[ANALYTICS_LIVE]

Telemetry Dashboard

Live tracking of cumulative scan count, total points earned, and most common materials deposited.



The Smart RVM aligns technical innovation with user incentives and global sustainability goals.



Acknowledging current limitations maps the path toward deployment resilience.

Current Technical Constraints



Single-Class Limitation

[SINGLE_CLASS_LIMIT]

Detects only one dominant object per frame (multi-object YOLOv8 planned).



Environmental Sensitivity

[ENV_SENSITIVITY]

Vulnerable to extreme lighting, poor camera angles, and complex background noise.



Infrastructure

[LOCAL_INFRASTRUCTURE]

Currently relies on a local Docker container without a cloud fallback protocol.

Ethical Considerations



Dataset Bias

[DATASET_BIAS]

Over-representation of Western market items may skew accuracy across different demographics.



Misclassification Risk

[MISCLASSIFICATION_RISK]

High environmental stakes if hazardous waste is erroneously routed to plastic streams.



Carbon Footprint

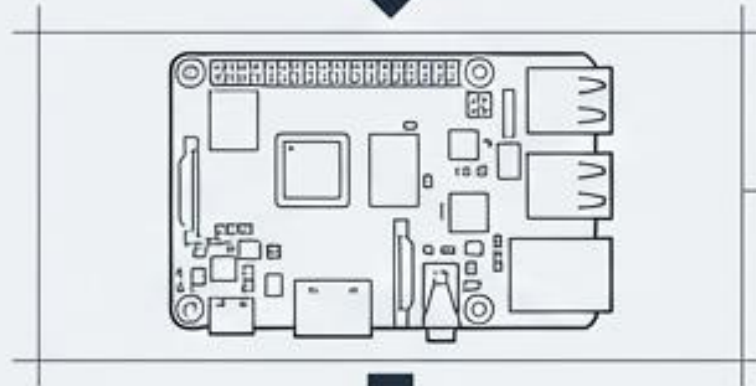
[CARBON_FOOTPRINT]

Heavy computational energy required for model training and deep learning execution.

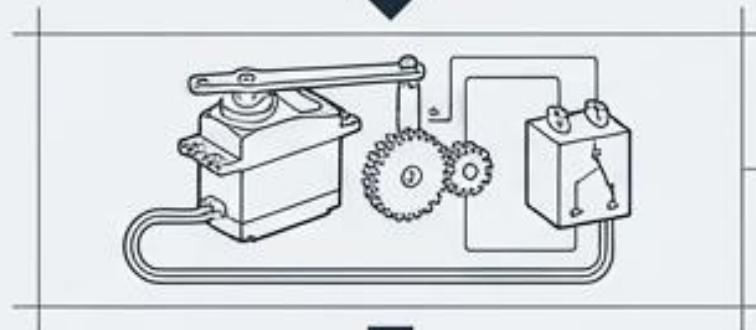
Translating digital ML predictions into mechanical hardware actuation.



The API Output **Payload**



The Edge Controller: Microcontroller parses the JSON via serial/network.



Physical Actuation: Controller drives servo motor mechanism.



The Clean Loop: Gate opens. Proximity/weight sensors trigger final user **reward**.

The roadmap scales the prototype from a web application to an Edge-AI physical machine.

